

Waves • Student Workbook

Name: _____ Class: _____

1 • Wave basics

1. Explain the statement: “waves transfer energy without transferring matter.”

2. Compare transverse and longitudinal waves. Include a labelled sketch.

2 • Equations

$$T = 1/f \quad v = f\lambda$$

3. A wave has $f = 8 \text{ Hz}$. Find T .

4. A wave has $f = 12 \text{ Hz}$ and $\lambda = 0.40 \text{ m}$. Find v .

3 • Ripple tank

5. Draw and explain reflection, refraction and diffraction of plane water waves.

6. Explain why a stroboscope is useful.

4 • Sound

7. Explain the bell-jar experiment and what it shows.

8. An echo returns after 0.60 s. Speed of sound = 340 m/s. Find the distance to the reflecting wall.

9. State how frequency and amplitude affect what we hear.

5 • EM waves and light

10. Write the seven EM regions in increasing wavelength.

11. Draw a reflection ray diagram and label normal, i and r.

12. State the two conditions for total internal reflection.

6 • Lenses

13. Complete an image-properties table for object beyond 2F, at 2F, between F and 2F, and inside F.

14. An image is 9 cm high and the object is 3 cm high. Find magnification.